Lab 8: Cellular Genetics — Mitosis and Meiosis

BIOL-1

Name: ______________________________	**Date:** ______________________________

Overview

Every cell in your body came from a single fertilized egg that divided over and over again. That process — **cell division** — is how you grow, repair wounds, and replace worn-out cells. But not all cell division is the same:

- **Mitosis** produces identical copies of a cell (for growth and repair).
- **Meiosis** produces unique sex cells called gametes (for reproduction).

In this lab, you will walk through both processes, compare them side by side, and discover how meiosis generates the extraordinary genetic diversity that makes every individual unique.

Learning Objectives

By the end of this lab, you will be able to:

1. Describe the stages of the cell cycle, including interphase and mitosis.
2. Compare what happens in each phase of mitosis vs. meiosis.
3. Explain why meiosis produces genetically unique cells.
4. Identify three sources of genetic variation in sexual reproduction.

Materials

- This worksheet
- Colored pencils (optional, for drawing)

Part 1: The Cell Cycle

Before a cell divides, it must prepare. Interphase is the "preparation phase" — the cell spends most of its life here.

1. Fill in what happens during each stage of interphase:

Stage	What Happens
G_1 (Gap 1)	
S (Synthesis)	
G_2 (Gap 2)	

2. After S phase, each chromosome consists of two identical copies joined at the centromere. What are these two copies called?

Part 2: Mitosis — Making Identical Copies

Mitosis is used for growth, repair, and replacing old cells. A skin cell divides by mitosis about every 2-3 weeks.

3. Fill in what happens during each phase of mitosis. Use the memory trick: PMAT.

Phase	What Happens	Memory Cue
Prophase		Chromosomes condense and become visible
Metaphase		Chromosomes line up in the **M**iddle
Anaphase		Chromosomes are pulled **A**part
Telophase		**T**wo new nuclei form

4. What separates during anaphase of mitosis — homologous chromosomes or sister chromatids?

5. After mitosis and cytokinesis, how many daughter cells are produced?

6. Are the daughter cells genetically identical to the parent cell?

7. Are the daughter cells diploid (2n) or haploid (n)?

8. How does cytokinesis differ between animal cells and plant cells?

Part 3: Meiosis — Making Unique Gametes

Meiosis happens only in the ovaries and testes. It produces egg and sperm cells with half the chromosome number, so that when they fuse at fertilization, the full number is restored.

9. How many rounds of division occur in meiosis?

10. How many daughter cells does meiosis produce from one parent cell?

11. Are the daughter cells diploid or haploid?

12. Are the daughter cells genetically identical to each other?

13. What separates during Anaphase I?

14. What separates during Anaphase II?

15. What is crossing over? During which phase does it happen? Why does it matter?

16. What is independent assortment? Why does random orientation of chromosomes at Metaphase I create diversity?

Part 4: Side-by-Side Comparison

17. Fill in the comparison table:

Feature	Mitosis	Meiosis
Number of divisions		
Number of daughter cells		
Ploidy of daughter cells		
Genetically identical to parent?		
Purpose		
Crossing over?		

Part 5: Genetic Variation and Real-World Connections

18. Name the three sources of genetic variation in sexual reproduction:

1.
2.
3.

19. In humans (n = 23), how many unique gamete combinations are possible from independent assortment alone? (Calculate 2^{23})

20. What is nondisjunction? Name one condition it can cause.

21. Using what you learned in this lab, explain in your own words why you are not genetically identical to your siblings (even though you have the same parents).